LOWER FRONT SUSPENSION BALL JOINT INSPECTION AND STEERING KNUCKLE REPLACEMENT

AFFECTED VEHICLES: CERTAIN 2000-2001 LINCOLN LS

OVERVIEW

The front suspension lower ball joint nuts on certain 2000-2001 Lincoln LS vehicles may not have been installed to the correct torque specification. To correctly identify these vehicles you will check the torque of the ball joint nuts. Depending on the results of the torque check, you will either retorque the nut(s) or replace the steering knuckle(s). This procedure will give detail for both the torque check and the steering knuckle replacement.

LOWER FRONT BALL JOINT NUT — TORQUE CHECK

1. Raise the vehicle on a hoist.
2. NOTE: Marking the wheel and a wheel stud will provide a reference for reassembly to maintain proper wheel balance. Place a mark on each front wheel and wheel stud. Remove the front wheel and tire assemblies.

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NOTE: If the vehicle has less than 15 miles, do not check the torque of the ball joint nuts. Instead, tighten the ball joint nuts to 150 Nm (111 lb-ft), then proceed to Step 4.

3. NOTE: You **MUST** use the hex holding feature of the ball joint stud to prevent the stud from turning while checking the torque of the nut. A suitable 21 mm crow foot wrench will be required to properly perform the torque check.

   Check the torque of both lower front ball joint nuts as follows. See Figure 1.

   A. If a beam or dial type torque wrench is used, apply force to the torque wrench until the nut begins to turn (Do not exceed 150 Nm [111 lb-ft]), record the torque.
   - If the torque is a minimum of 80 Nm (59 lb-ft), retorque the nut(s) to 150 Nm (111 lb-ft).
   - If the torque is less than 80 Nm (59 lb-ft), the steering knuckle must be replaced.

   B. If a "click" type torque wrench is used, set the wrench to 80 Nm (59 lb-ft), then apply force to the torque wrench until the nut begins to turn **OR** the wrench "clicks".
   - If the nut does not turn before the wrench "clicks", retorque the nut(s) to 150 Nm (111 lb-ft).
   - If the nut turns before the wrench "clicks", the steering knuckle must be replaced.

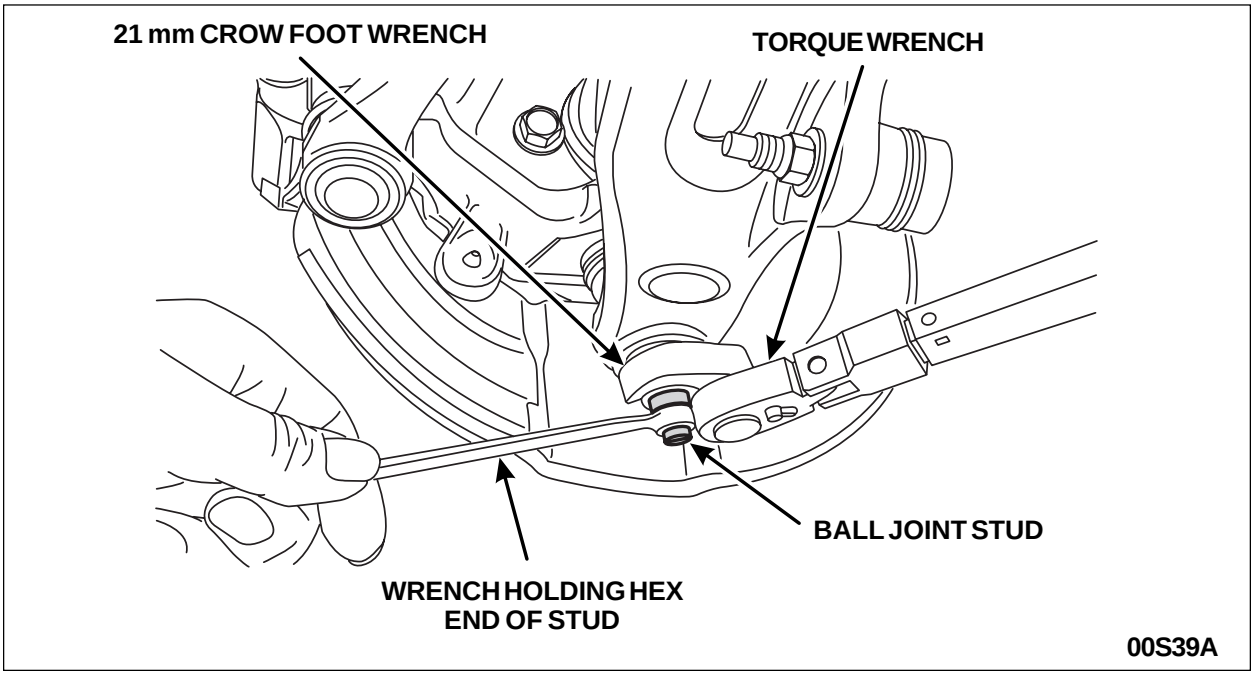


FIGURE 1

4. On the side(s) of the vehicle that do not require steering knuckle replacement, install the wheel and tire assembly aligning the marks on the wheel and wheel stud. Tighten to 90 Nm (66 lb-ft).

5. If neither knuckle requires replacement, lower the vehicle. No further repair is necessary.

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STEERING KNUCKLE-REMOVAL

1. **CAUTION: Do not allow the caliper anchor plate/brake assembly to hang from the brake hose.**

   Remove and discard the caliper anchor plate bolts, then lift the caliper anchor plate/brake assembly from the brake disc. Support the assembly with mechanics wire. See Figure 2.

2. NOTE: Marking the brake disc and a wheel stud will provide a reference for correct reassembly and maintain the lowest brake disc runout.

   Mark the brake disc and one wheel stud. See Figure 2.

3. Remove and discard the tinnerman nuts (if equipped) from the wheel studs and remove the brake disc. See Figure 2.

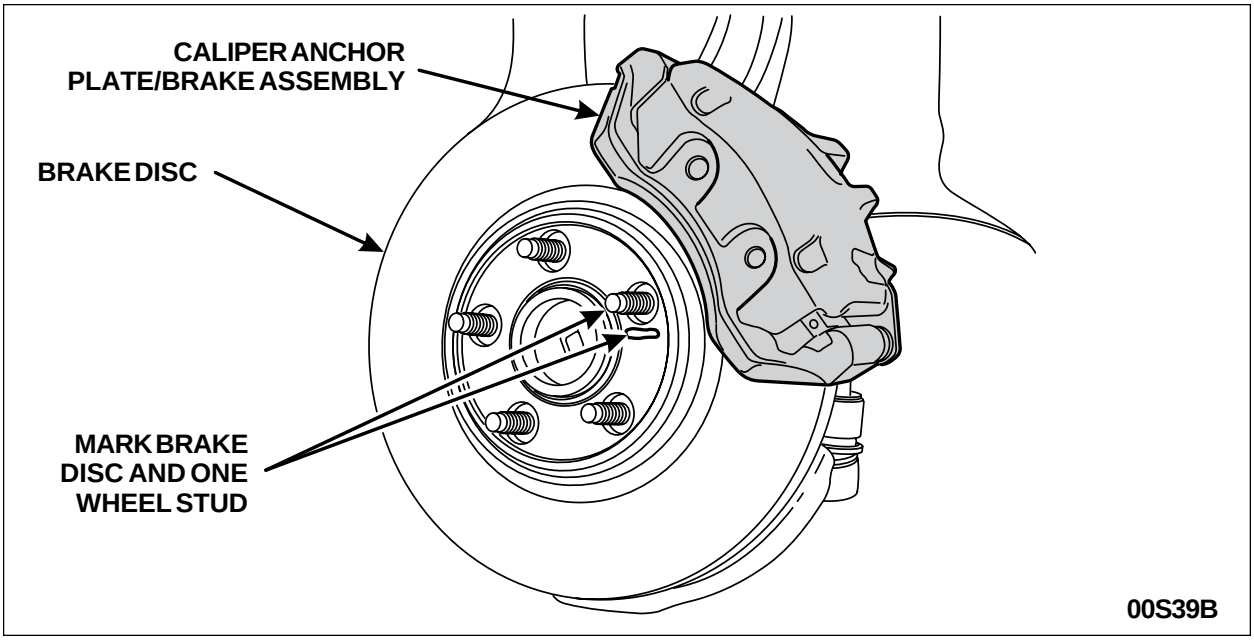


FIGURE 2

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4. Remove the three pushpins from the inner fender skirt, then position the skirt aside to access the anti-lock brake (ABS) sensor electrical connector. See Figure 3.

5. **CAUTION: Do not remove the ABS sensor and wire from the hub assembly. The sensor O-ring may be damaged.**

   Disconnect the ABS sensor from the vehicle wiring harness. Detach the ABS sensor wiring from the retaining clips. See Figure 3.

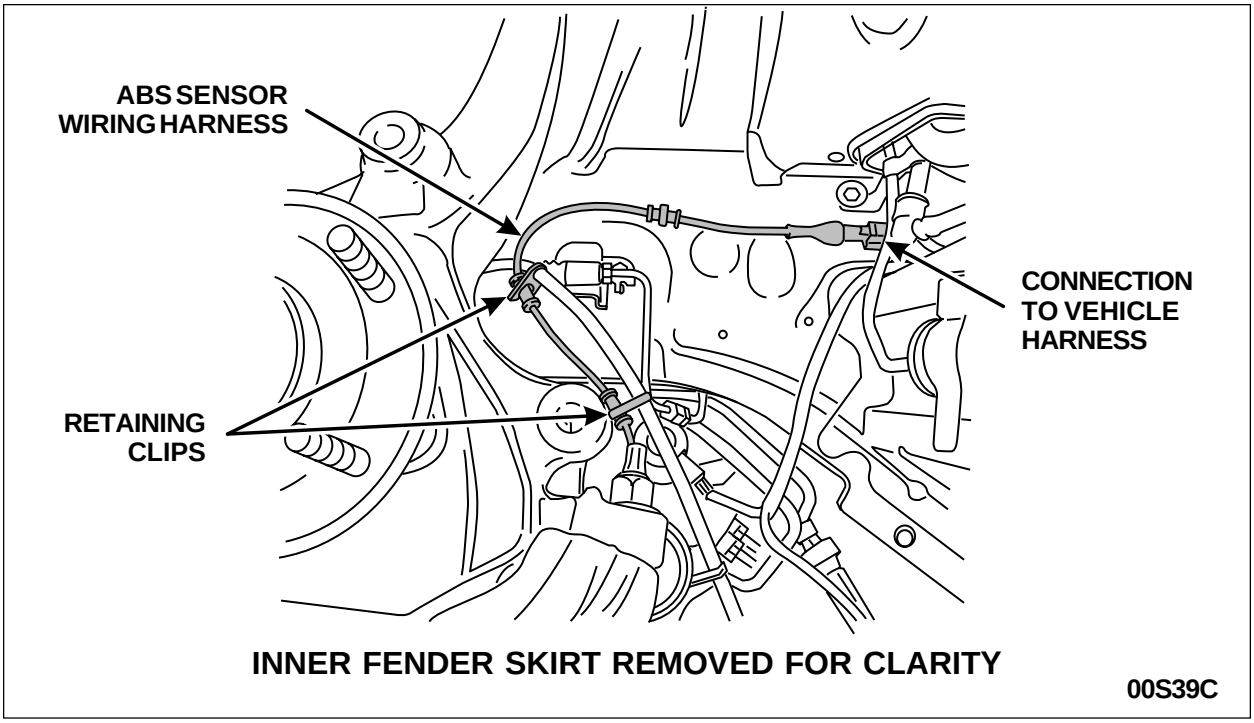


FIGURE 3

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6. **CAUTION: The wheel hub assembly is not pressed into the knuckle. Do not use a slide hammer or strike the back of the wheel hub assembly.**

   NOTE: If necessary, remove any rust or corrosion from the back of the wheel hub assembly. Apply Rust Penetrant and Inhibitor (D7AZ-19A501-AA) or equivalent to the seam between the wheel hub assembly and the bore of the steering knuckle.

   Remove and discard the wheel hub assembly bolts. Remove the wheel hub assembly. See Figure 4.

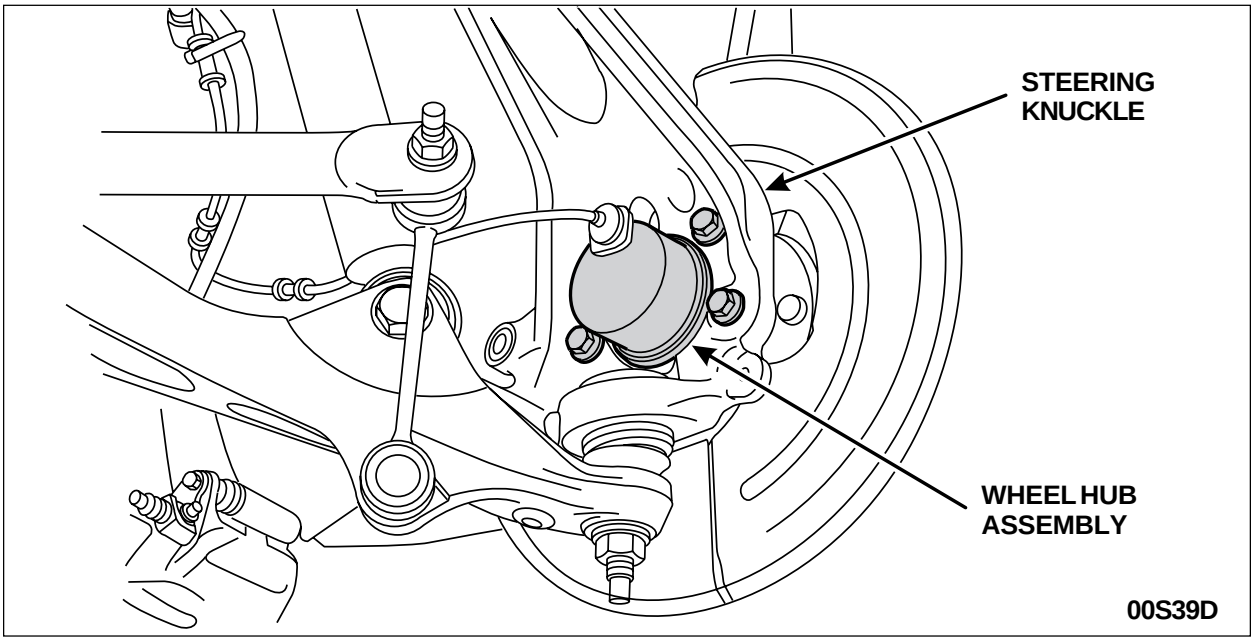


FIGURE 4

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7. NOTE: Use the hex holding feature of the tie rod end stud to prevent the stud from turning while removing the nut. Remove and discard the tie rod end nut. Separate the tie rod end from the steering knuckle. See Figure 5.
8. NOTE: Use the hex holding feature of the ball joint stud to prevent the stud from turning while loosening the nut.

   Loosen the upper ball joint nut. Do not remove the nut at this time. See Figure 5.
9. NOTE: Use the hex holding feature of the ball joint stud to prevent the stud from turning while removing the nut.

   Remove and discard the lower ball joint nut. See Figure 5.
10. **CAUTION: The upper control arm will be damaged if the steering knuckle is allowed to hang from the upper control arm.**

    Lift and separate the steering knuckle from the lower control arm. Remove and discard the upper ball joint nut, then remove and discard the steering knuckle. See Figure 5.

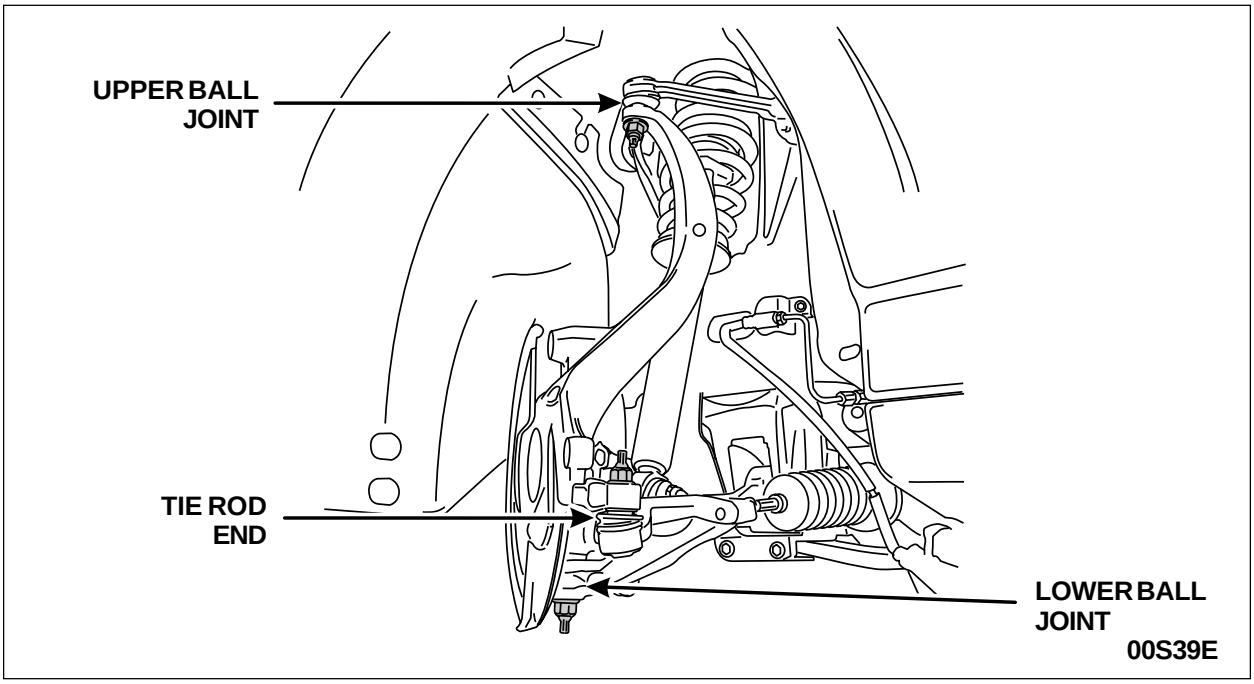


FIGURE 5

STEERING KNUCKLE-INSTALLATION

NOTE: For a correct installation, all of the ***new*** fasteners included in the kit must be used. Do not use power tools for installing fasteners.

1. Position the lower ball joint stud of the ***new*** steering knuckle into the lower control arm. Install a ***new*** nut on the ball joint. Do not tighten nut at this time.
2. Install the upper ball joint stud into the steering knuckle. Install a ***new*** nut onto the ball joint. Tighten the nut to 90 Nm (66 ft-lb).

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3. Tighten the lower ball joint nut to 150 Nm (111 lb-ft).
4. Install the tie rod end into the steering knuckle. Install a ***new*** nut on the tie rod end. Tighten the nut to 100 Nm (74 lb-ft).
5. Install the brake disc dust shield with ***new*** rivets.
6. Apply a light coat of Motorcraft High Temperature Nickel Anti-Seize Lubricant (F6AZ-9L494-AA) to the outer surface of the wheel hub assembly that contacts the bore of the steering knuckle.
7. Install the wheel hub assembly into the steering knuckle using ***new*** bolts. Tighten the bolts to 90 Nm (66 lb-ft).
8. Route the ABS sensor wiring and attach the wiring to the retaining clips. Connect the ABS sensor wiring to the vehicle harness.
9. Position the inner fender skirt and install the pushpins.
10. Install the brake disc, aligning the marks on the disc and the wheel stud.
11. Install the caliper anchor plate/brake assembly onto the brake disc. Install ***new*** anchor plate bolts and tighten to 103 Nm (76 lb-ft).
12. Install the wheel and tire assembly, aligning the marks on the wheel and wheel stud. Tighten to 90 Nm (66 lb-ft).
13. Lower the vehicle.
14. Check the toe-in and adjust as necessary.

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